

Technical Document #73: Blockchain - Comprehensive Overview

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Blockchain is a distributed ledger technology that records transactions across many computers. It ensures transparency and immutability.

Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions. It is energy-intensive but secure.

Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Smart contracts are self-executing contracts with terms directly written into code.
- Cross-chain technology enables communication between different blockchains.
- Non-fungible tokens (NFTs) represent unique digital assets on the blockchain.